

O-SIGHT: LOCAL-FIRST FINANCIAL INTELLIGENCE DASHBOARD

OPay National Innovation Challenge Application 2026

1. Executive Summary

In Nigeria's current high-inflation environment, traditional budgeting has failed. While millions of Nigerians conduct daily transactions using digital wallets like OPay, they struggle with "statement blindness," unable to track where their money is going.

O-Sight is a privacy-first, local-first financial intelligence dashboard. It allows users to export and drag-and-drop their raw statement files (specifically optimized for OPay Excel exports). The application automatically runs a Python-based pipeline that cleans, reconciles, and evaluates transactions to render an interactive web dashboard. By providing automated audits of spending trends, savings progress, and hidden bank charges, O-Sight empowers users to take control of their finances securely.

2. The Problem

Modern fintech apps have made transferring money seamless, but monitoring spending remains frustrating. Modern fintech apps have made transferring money seamless, but analyzing transaction logs remains highly frustrating:

- a) **Cryptic Transaction Narration & Merchant Fragmentation:** Statements are littered with unreadable descriptions, listing raw account fragments, USSD codes, or labels like "Transfer to 8102..." or "Third-Party Merchant Order". Furthermore, the same merchant often appears under slightly different names (e.g., "MTN Airtime", "MTN DATA", and "MTN TopUp"), bloating lists and preventing accurate merchant categorization.
- b) **The Security and Trust Gap:** Existing budgeting software requires users to link their live bank accounts using open banking APIs or credential sharing. Because of security concerns, high fraud rates, and unstable banking networks in Nigeria, users are highly reluctant to link their credentials to third-party applications.
- c) **Sweeps and Mismatches (Savings bloat):** Generic tracking tools treat all outgoing transfers as spending. However, a significant portion of a user's outflow consists of moving money to savings wallets or internal accounts (e.g., transfers to OPay's *OWealth*, *Spend & Save*, *Fixed Deposit*, *Targets Deposit*, or external accounts like Cowrywise). This bloats "spending" metrics and distorts the user's true net savings rate and income-to-spending ratio.

3. Proposed Solution & Architecture

O-Sight is designed as a local-first, containerized application built on a modern stack: a React/Vite frontend and a FastAPI/Pandas backend. Instead of requiring live bank integrations, the user simply exports their OPay transaction Excel sheet and drags it into the dashboard.

The core technology pipeline operates through four main phases:

- a) **Data Ingestion and Reconciliation (opay.py):** Upon file upload, the OPayAnalyzer reads the header to extract the user's account details (Account Name, Type, Number, and Period) and the bank's reported totals. It parses the transaction logs, converts currency

symbols (~~N~~) to float values, and reconciles the calculations. It flags inflow/outflow discrepancies if the computed sums mismatch the reported totals by more than 0.01 Naira.

- b) Fuzzy NLP Cleanups and Merchant Deduplication (nlp.py): To resolve fragmented names (e.g., consolidating "BOLT RIDES", "BOLT GIG", and "Bolt"), the backend runs a two-step cleaning process:
 - i. `clean_counterparty`: Normalizes raw text (lowercasing, stripping special characters, and sorting name tokens alphabetically to guarantee "FIRST LAST" maps identically to "LAST FIRST").
 - ii. `fuzzy_deduplicate`: Leverages Python's `SequenceMatcher` to evaluate text similarity. Any counterparties with a similarity ratio above 0.85 (or 0.80 for parent entities) are grouped under a single canonical name, resolving "Evil Twins" fragmentation.
- c) Hybrid Mask Resolution (intelligence.py): Fintech statements often mask account numbers (e.g., "123***789"). The engine builds an in-memory registry of full, unmasked account numbers mapped to specific counterparties. When it encounters a masked account, it matches the unmasked prefix (first 3 digits), suffix (last 3 digits), and counterparty name against the registry to restore the complete account details.
- d) Rule-Based Savings Filtration (opay.py): To eliminate savings bloating, the parser scans description strings for keywords like "OWealth", "Spend & Save", "Fixed Deposit", and "Targets Deposit". It isolates these transactions into "Savings & Investments" or "Savings Withdrawal" categories, allowing the intelligence engine to extract the user's true outflow and savings rate.
- e) Weighted Financial Health Score v2 (intelligence.py): The engine scores the user's financial profile out of 100 points, using seven weighted indicators:
 - i. *Savings Rate (25% weight)*: Evaluates net savings relative to real inflow.
 - ii. *Income Stability (20% weight)*: Measures the coefficient of variation (CV) of monthly inflows.
 - iii. *Spending Consistency (15% weight)*: Evaluates volatility in monthly expenditures.
 - iv. *Bank Charges Ratio (10% weight)*: Tracks the percentage of outflows spent on USSD/bank charges.
 - v. *Balance Trend (15% weight)*: Evaluates if average balances are growing, stable, or declining.
 - vi. *Anomaly Frequency (5% weight)*: Flags unusual spending spikes.
 - vii. *Diversification (10% weight)*: Measures spending concentration across categories.
- f) Zero-Retention Data Privacy (main.py): For security, statements are uploaded via FastAPI to temporary files (tempfile.mkstemp). The data is analyzed in memory, and the temporary file is immediately unlinked and deleted in a finally block before the JSON response is

returned to the browser. The frontend stores this state only in React's active runtime memory, leaving zero footprints on any server database.

4. Target Market & Economic Relevance

- a) Primary Target Audience: O-Sight's core target market is the growing population of digitally active Nigerians (aged 18–35), specifically:
 - i. Students & Young Professionals: Heavy mobile money users who conduct multiple micro-transactions (e.g., transfers, airtime purchases) daily and need a simple, self-auditing tool to curb impulse spending.
 - ii. Freelancers & Gig Workers: Individuals who receive multiple unstructured payments from different clients and need to separate their actual business income from personal sweeps and savings withdrawals.
 - iii. Micro-Merchants: Small retail vendors who handle a large volume of customer transfers via personal accounts and require a automated tool to check daily sales velocity and balance sheets.
- b) Social and Economic Impact in Nigeria:
 - i. Combating Transaction Inflation: With inflation and rising costs in Nigeria, users need to manage their cash flows down to the single Kobo. O-Sight makes hidden transaction charges such as electronic transfer levies, SMS alerts, and USSD fees—highly visible. This visibility drives users to adopt cost-saving transaction behaviors.
 - ii. Democratizing Financial Literacy: Rather than manual budgeting apps that demand strict discipline and typing every single transaction, O-Sight handles 100% of the extraction and categorizing automatically. This reduces the friction to learning healthy financial habits.
 - iii. Data Privacy as a Trust Builder: Traditional open banking solutions fail to gain traction in Nigeria due to safety fears. By showing users that their financial statements are processed locally and never stored on a database, O-Sight establishes a high trust standard, encouraging wider adoption of personal finance tools.
 - iv. Alignment with Digital Economy Goals: O-Sight aligns with Nigeria's National Digital Economy Policy and Strategy (NDEPS), specifically on driving digital literacy, financial inclusion, and data-backed economic empowerment for the youth.